Let the vectors *OB*, *OC* and *OD* represent E_R, E_L and E_C. If $E_L > E_C$ the resultant of these two is $(E_L - E_C)$. This is represented by *OE*.

$$OE = OC - OD$$

$$= I_0\left(L\omega - \frac{1}{C\omega}\right); \text{ where } L\omega > \frac{1}{C\omega}$$

$$E_0 = [OB^2 + BF^2]^{\frac{1}{2}} = [OB^2 + OE^2]^{\frac{1}{2}}$$

$$= \left[I_0^2R^2 + I_0^2\left(L\omega - \frac{1}{C\omega}\right)^2\right]^{\frac{1}{2}}$$

$$= I_0\left[\left(L\omega - \frac{1}{C\omega}\right)^2 + R^2\right]^{\frac{1}{2}}$$

$$I_0 = \frac{E_0}{\sqrt{\left[\left(L\omega - \frac{1}{C\omega}\right)^2 + R^2\right]}}$$

The current lags behind the applied voltage by ϕ given as

$$\phi = \tan^{-1}\left(\frac{L\omega - 1/C\omega}{R}\right)$$

j operator method

Use of operator j in study of A.C. Circuits

The operator *j* is defined as a quantity which is numerically equal to $\sqrt{(-1)}$ and which represents the rotation of a vector through 90° in anticlockwise direction, $-j$ represents the rotation of a vector through 90° in clockwise direction. The above facts about *j* are helpful in studying the A.C. circuits. We know that in A.C. circuits, E_L and E_C always lie at 90° in anticlockwise and clockwise direction respectively with respect to E_R (Fig. 13.20). Hence total emf of a circuit having *L*, *C*, *R* will be

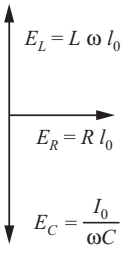


FIG. 13.20

$$E = E_R + jE_L - jE_C$$

Ordinarily a source of alternating e.m.f. *E* is denoted by E_0 sin ωt.

This is actually the imaginary part of the complex form of alternating e.m.f.,

$$E = E_0 e^{j\omega t}$$

The instantaneous current in the A.C. circuit has been expressed as $I = I_0$ sin $(\omega t - \phi)$. This expression is the imaginary part of the complex current given by

$$I = I_0\, e^{j(\omega t - \phi)}$$

Since the voltage across the inductor leads the current passing through it by 90°, the inductive reactance ω*L* can be written as $j\omega L$. $Z_L = j\omega L = jX_L$

Since the voltage across the capacitor lags the current passing through it by 90°, the capacitive reactance $1/\omega C$ can be written as

$$-j/\omega C = 1/j\omega C.\ Z_c = \frac{1}{j\omega C} = -\frac{j}{\omega C} = -jX_C$$

A complex impedance can be written as the sum of a real term and imaginary term which are to be called resistance and complex reactance respectively,

$$Z = R + jX$$

where $X = X_L - X_C$ is the effective reactance in the circuit.

LCR Circuit (Series Resonance Circuit) [Second Method]

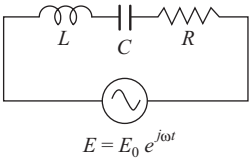


FIG. 13.21

Consider a circuit containing an inductance L, a capacitance C and a resistance R joined in series. This series circuit is connected to an AC supply given by

$$E = E_0 e^{j\omega t} \quad \text{(Fig. 13.21)} \qquad ...(1)$$

The total complex impedance is

$$Z = Z_R + Z_L + Z_C$$

$$= R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$= \sqrt{R^2 + (\omega L - 1/\omega C)^2}\; e^{j\phi}$$

where $\tan\phi = \dfrac{(\omega L - 1/\omega C)}{R}$ ...(2)

Using Ohm's law in complex form, the 'complex' current in the circuit is

$$I = \frac{E}{Z} = \frac{E_0 e^{j\omega t}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}\; e^{j\phi}}$$

$$\therefore \quad I = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}\, e^{j(\omega t - \phi)} \qquad ...(3)$$

But $$I_0 = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\therefore \quad I = I_0 e^{j(\omega t - \phi)} \qquad ...(4)$$

The actual emf is the imaginary part of the equivalent complex emf. Hence the actual current in the circuit is obtained by taking the imaginary part of the above 'complex' current.

$$\therefore \quad i = Im,\,(I) = \frac{E_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}\, \sin(\omega t - \phi) \qquad ...(5)$$

The equivalent impedance of the series LCR circuit is

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

The current 'lags' behind the voltage by an angle

$$\phi = \tan^{-1}\frac{(\omega L - 1/\omega C)}{R}$$